

Galerkin / Least-square

$$Re U_x - U_{xx} = 0$$

$$U(0) = 0 \rightarrow U(1) = 1$$

Solution

$$R = Re U_x - U_{xx} \rightarrow W_i = FN_i + \gamma L(N_i) \quad L(1) = Re \frac{\partial 1}{\partial x} - \frac{\partial^2 1}{\partial x^2}$$

$$\rightarrow \int_0^1 R W_i = \int_0^1 (Re U_x - U_{xx})(FN_i + \gamma L(N_i))$$

$$= \int_0^1 Re U_x (FN_i + \gamma L(N_i)) - \int_0^1 U_{xx} (FN_i + \gamma L(N_i))$$

$$\rightarrow L(N_i) = Re \frac{\partial N_i}{\partial x} - \frac{\partial^2 (N_i)}{\partial x^2} = Re N_i'$$

$$\rightarrow \int_0^1 Re U_x FN_i + Re^2 U_x \gamma N_i' - \left[U_x FN_i + U_x \gamma RN_i' \right] - \int_0^1 U_{xx} FN_i + U_{xx} \gamma RN_i' = 0$$

$$\rightarrow \int_0^1 Re U_x FN_i + U_x N_i' (Re^2 \gamma + F) - \left[U_x(1)F + N_i U_x(1) \gamma R - U_x(0)F - U_x(0) \gamma RN_i' \right] = 0$$

$$U_x = U_j N_j'$$

$$\rightarrow U_j \left[\int_{x_1}^{x_2} Re F N_j' N_i + (Re^2 \gamma + F) N_j' N_i \right] = 0$$

$$N_1' = \frac{-1}{L_e} \rightarrow N_2' = \frac{1}{L_e}$$

$$\int_{x_1}^{x_2} N_i \cdot dx = \int_{x_2}^{x_3} N_i \cdot dx = \frac{L_e}{2}$$

$$K_{11} = Re F N_1' \int_{x_2}^{x_3} N_1 \cdot dx + (Re^2 \gamma + F) N_1'^2 \int_{x_2}^{x_3} dx = \frac{-Re F}{2} + \frac{Re^2 \gamma + F}{L_e}$$

$$K_{22} = Re F N_2' \int_{x_2}^{x_3} N_2 \cdot dx + (Re^2 \gamma + F) N_2'^2 \int_{x_2}^{x_3} dx = \frac{Re F}{2} + \frac{Re^2 \gamma + F}{L_e}$$

$$K_{12} = Re F N_2' \int_{x_2}^{x_3} N_1 \cdot dx + (Re^2 \gamma + F) N_2' N_1' \int_{x_2}^{x_3} dx = \frac{Re F}{2} - \frac{(Re^2 \gamma + F)}{L_e}$$

$$K_{21} = Re F N_1' \int_{x_2}^{x_3} N_2 \cdot dx + (Re^2 \gamma + F) N_1' N_2' \int_{x_2}^{x_3} dx = \frac{-Re F}{2} - \frac{(Re^2 \gamma + F)}{L_e}$$

For LS $\rightarrow (F=0 \wedge \gamma=1)$

$$\text{Local Matrix} \rightarrow \begin{bmatrix} \frac{R_e^2}{L_e} & -\frac{R_e^2}{L_e} \\ -\frac{R_e^2}{L_e} & \frac{R_e^2}{L_e} \end{bmatrix}$$

for SG $\rightarrow (F=1 \wedge \gamma=0)$

$$\text{Local Matrix} \rightarrow \begin{bmatrix} -\frac{R_e}{2} + \frac{1}{L_e} & \frac{R_e}{2} - \frac{1}{L_e} \\ -\frac{R_e}{2} - \frac{1}{L_e} & \frac{R_e}{2} + \frac{1}{L_e} \end{bmatrix}$$

for GLS $\rightarrow (F=1)$

$$\text{Local Matrix} \rightarrow \begin{bmatrix} -\frac{R_e}{2} + \frac{R_e^2 \gamma + 1}{L_e} & \frac{R_e}{2} - \frac{(R_e^2 \gamma + 1)}{L_e} \\ -\frac{R_e}{2} - \frac{(R_e^2 \gamma + 1)}{L_e} & \frac{R_e}{2} + \frac{(R_e^2 \gamma + 1)}{L_e} \end{bmatrix}$$

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$$\gamma = \frac{1}{\left| \left(\frac{2R_e}{L_e} \right)^2 + \left(\frac{4n-1}{L_e^2} \right)^2 \right|}$$

Assignment $\neq$

$$\rho_e \frac{\partial T}{\partial x} - \frac{\partial^2 T}{\partial x^2} = 0$$

$$\rho_e = \frac{\mu L}{\alpha}$$

$$T(0) = 0 \wedge T(1) = 1$$

Same as Lecture example with $T \equiv u \quad \rho_e \equiv R_e$